Send Orders for Reprints to reprints@benthamscience.net

RESEARCH ARTICLE

Integrating IoMT and Federated Learning for Advanced Healthcare Monitoring in Healthcare 5.0

Bassam Almogadwy[1,*] and Abdulrahman Alqarafi[1]

[1]*College of Computer Science and Engineering, Taibah University, Medina, 42353, Saudi Arabia*

© 2025 The Author(s). Published by Bentham Science Publisher. This is an open access article published under CC BY 4.0 https://creativecommons.org/licenses/by/4.0/legalcode

ARTICLE HISTORY

Received: February 05, 2025
Revised: April 23, 2025
Accepted: May 02, 2025

DOI:
10.2174/0113892010380715250728143022

Abstract: ***Introduction*:** The Internet of Medical Things (IoMT) has made it possible to create advanced health monitoring systems. It allows the system to detect problems early, thereby mitigating long-term effects. This development will likely enhance the quality of healthcare professionals by reducing their workload and healthcare costs. The IoT in medical technology offers a wide range of information technology capabilities, including intelligent and collaborative healthcare solutions. Aggregating health data in a single repository raises security, copyright, and compliance issues when building a complex machine-learning model.

***Method*:** Federated learning overcomes the above challenges by dispersing a global learning model through a central aggregate server. It retains mastery of patient data in a local participant who ensures data privacy and integrity. This research aims to develop an advanced healthcare monitoring system utilizing federated learning techniques. The system is designed to enable healthcare providers to effectively track patient health through medical sensors and respond promptly when necessary.

***Results*:** The federated learning-based XGBoost model achieved a predictive accuracy of 97.2% in diagnosing Parkinson's disease. Additionally, the system demonstrated improved privacy preservation, significantly reducing sensitive data exposure with minimal computational overhead, confirming its practical effectiveness in clinical scenarios.

***Discussion*:** By leveraging federated learning, the proposed approach seeks to enhance the efficiency and effectiveness of health monitoring in clinical settings. To achieve accurate classification and early detection of Parkinson's disease, the study employs two key machine learning algorithms: Support Vector Machine (SVM) and Extreme Gradient Boosting (XGBoost). These methods were selected for their statistical robustness and suitability for the task at hand.

***Conclusion*:** The combination of federated learning, SVM, and XGBoost enhances healthcare monitoring and ensures patient data privacy and integrity.

Keywords: Healthcare 5.0, Federated learning, artificial intelligence, IoMT.

1. INTRODUCTION

The Internet of Medical Things (IoMT) is a rapidly evolving technology that connects medical devices and applications via the Internet, enabling real-time monitoring and enhanced patient care [1]. Many things can be connected via IoT (the Internet of Things) to collect data to improve human well-being, productivity, and efficiency [2-4]. The notions of smart cities, smart grids, and smart homes have been around for a while, and they are gradually making their way into our everyday lives [5-7]. Utilizing an Internet of Things-based monitoring system of patient health is one of the most promising new technology options for closing the gap in health equity that exists around the globe [8]. The Internet of Medical Things is another term for various technologies that are part of the IoT. (IoMT). Even though this study will concentrate on the medical field, the words "Internet of Things" and "Internet of Medical Things" will be used interchangeably during this investigation. In addition, the Internet of Things can bring about substantial advancements in the fields of healthcare and public safety [9]. Connecting one's body to the Internet enables one to get insight into their habits, health, and the quality of their environment. This paves the way for real-time, remote health monitoring by medical professionals [10-13]. In addition, the collected information might be utilized to advocate for the implementation of ill-

*Address correspondence to this author at the College of Computer Science and Engineering, Taibah University, Medina 42353, Saudi Arabia;
E-mail: bmogadwi@taibahu.edu.sa

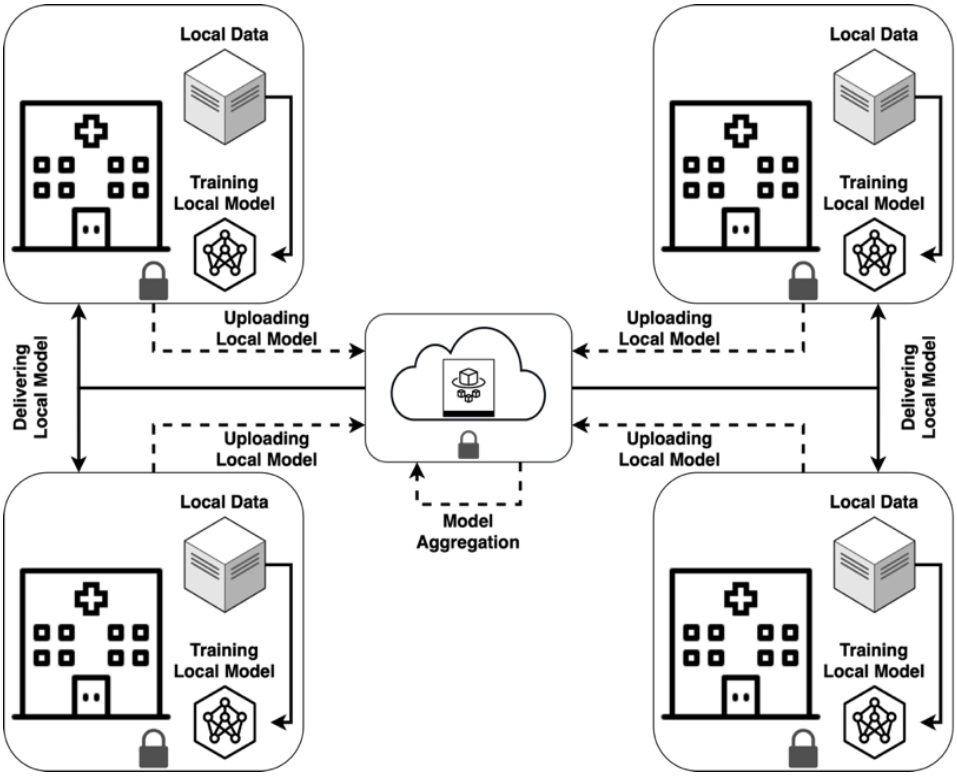

Fig. (1). A federated learning technique in healthcare system. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

ness, trauma, protection, early diagnosis, and rehabilitation programs supported by scientific evidence [14].

In today's environment, facilitating patient transfers to healthcare facilities for routine checkups presents significant challenges, including limited mobility, transportation accessibility, and rising healthcare costs, all of which can hinder timely and effective medical care [15-18]. Additional hurdles include waiting in long lines, punctuality issues, and the risk of illness from polluted air. Hence, the health industry has been paying more attention to the healthcare services that can be delivered in the comfort of one's home, so that the patient can do other things without any uncertainty regarding his or her treatment. Advanced health monitoring technologies are being developed so that patients staying in remote places can benefit from the availability of professionals in urban areas. With this technology, which is a communication between the vital signs of a patient and the application through wireless transmission of collected data, the patient's safety and sound health are maintained because this device continuously monitors the vital signs [19-22].

It is of the utmost importance to develop efficient mining algorithms to evaluate medical information to contribute to the discovery of diseases, provide medical therapy, and improve patients' health [23-26]. ML methods have found applications in a wide number of fields, including image identification, language processing, and health care, among others [27]. As can be judged from the working of health care, where life or death is a matter of accuracy for a patient, in the pursuit of acquiring high precision in results, a large training database is needed for machine learning models [28-30]. In most cases, centralized training methods obtain huge amounts of data from a strong cloud server, which may pose a huge risk to consumer privacy, especially in the health sector [31-33].

Federated learning, or FL, represents a decentralized framework for machine learning in the context of the Internet of Things that allows multiple devices to collaborate on developing machine learning models without exposing their real data. This method reduces the leakage of patient information to unauthorized people and increases the effectiveness of intelligent healthcare systems [34-37]. As illustrated in Fig. (**1**), in the federated learning-based system of health care, devices collect health information from medical practitioners, and various IoT platforms work together for federated learning methods; machine learning techniques are used for assessing patient health, evaluating and, if necessary, initiating immediate support from the cloud. The very high level of security of federated learning in the analysis of distributed sensitive data was yet another key factor contributing to its rising popularity lately as a methodological framework. Instead of collecting data from many sites or following the approach of find-then-replicate, on a central server, a global model is trained, yet individual companies' data is kept separately. This approach, which is called "federated learning," whereby data from multiple sites is combined to form a single global model, precisely describes the working mechanism. This means producing a global model by summarizing data coming from sources with learning. This is singular as it happens without any sharing of information. The result of this procedure is the boosting of accuracy for machine learning algorithms. In this methodology,

privacy for personal patient data is kept intact in each healthcare setting [38].

Federated Learning (FL) is a decentralized machine learning technique that enables multiple institutions to collaboratively train a global model without sharing their local datasets, thus preserving data privacy and security. XGBoost (Extreme Gradient Boosting) is an efficient and scalable tree-based machine learning algorithm known for its high performance and predictive accuracy, widely utilized in healthcare analytics due to its robustness and interpretability.

Healthcare 5.0 [39-41] is a decentralized system that requires the main network architecture for linking healthcare devices to be fifth-generation connectivity. The Internet of Things (IoT) will generate data that artificial intelligence (AI) will be able to use. This will help the growth of digital health by concentrating not only on the health and well-being of individual patients but also on the quality of life and well-being of individuals located all over the world. The most important challenges in Healthcare 4.0 are the flow of information without any errors and with as little information loss as possible. The healthcare industry is undergoing a significant transformation with the advent of Healthcare 5.0, driven by automation and artificial intelligence (AI). These technologies are set to reshape various aspects of the healthcare workforce. Healthcare 5.0 introduces advanced capabilities such as precise disease prediction, remote patient monitoring, intelligent robotic systems, and innovative therapeutic approaches like virtual counseling for anxiety patients. AI is a general term that consists of many different kinds of machine intelligence. It means the ability to predict the outcome of the machine learning algorithm without human oversight. The recent technological advancements in the field have enabled the establishment of "smart health systems," integrating these high technologies into improved patient outcomes and better healthcare services.

To overcome the aforementioned issues, this work combines FL and advanced encryption to create a healthcare 5.0 system that is secure and protects patient privacy. The following is a list of the most important contributions that can be attributed to this study;

- The proposed model tries to include a variety of medical institutions. Pooling the locally trained models from various healthcare organizations may significantly enhance the system's overall performance and capabilities. This would leverage the combined knowledge and data from the sources, potentially making their solutions more reliable and holistic for the Health Care 5.0 framework.
- Federated learning is how the design and implementation of learning in Healthcare 5.0 can be done to acquire medical knowledge seamlessly and faster. This approach uses models trained locally from different institutions to enable rapid learning without exposing data to anyone, and the system can quickly assimilate and apply insights from a broad spectrum of medical data towards potential faster advancements in patient care and treatment strategies.
- To create an intelligent hybrid strategy to improve secure communication and the efficiency of healthcare monitoring.
- To propose an FL-based framework for healthcare 5.0, that not only develops an effective collaborative architecture using many IoMT gadgets, nevertheless also manages the whole training process.
- The suggested approach secures the Federated Learning (FL)--based healthcare 5.0 System by precisely evaluating its dependability on the critical security objectives of confidentiality, authenticity, and usability. This study aims to demonstrate that the operating costs created by our technique are insignificant when contrasted to the benefits it delivers in terms of both security and privacy.

This study presents a novel approach to integrating machine learning and federated learning techniques for enhanced healthcare monitoring within an IoMT framework. The key contributions of our work include:

1. A novel federated learning-based framework that ensures secure and decentralized healthcare data processing.
2. Enhanced privacy-preserving mechanisms to mitigate data security concerns in IoMT-based healthcare.
3. Comprehensive evaluation using additional performance metrics such as precision, recall, and F1-score.
4. Comparison with recent studies (2022–2024) to validate our model's effectiveness.

Cost-benefit analysis considering human efforts for practical implementation.

The remaining part of this article consists of the sections listed below. The second section offers overviews of relevant studies. The third section describes the recommended technique. The simulation and findings of the recommended strategy are presented in Section 4. Section 5 concludes with a discussion of the study's findings.

2. LITERATURE REVIEW

X. Du *et al*. [42] proposed a method that set out to do several things: look into how blockchain technology is being used in smart healthcare; create a unified theory of smart healthcare; specify blockchain's impact on smart healthcare; and, finally, design an advancement application framework for smart healthcare that takes stakeholder feedback into account. Ihnaini *et al*. [43] proposed a strategy for predicting the likelihood of developing diabetes that makes use of machine learning techniques. The suggested method can improve the suggested system's efficacy in properly forecasting and advising this life-threatening situation by merging information, which reduces unnecessary pressure on the system's computational resources. At last, a model for predicting the onset of diabetes is developed using an ensemble machine-learning strategy.

Data may be readily sent over numerous networks in today's world, which enables professionals and businesses to maximize their use of available resources while simultaneously satisfying the needs of society in the field of medicine.

Because of the Internet of Things, users now have access to healthcare services that are both reliable and highly effective. Intelligent sensors revolutionize the health sector by allowing communities to gain precise insight into their medical needs. Wearables have demonstrated their influence in the sense that they measure several physiological functions that can be used as guides to personal care. Health practitioners ensure the most effective treatment of individuals by monitoring various body systems. Data accumulated from these sensors can therefore be processed for reliable predictions of diseases and proactive health interventions [44]. Khan *et al*. [45] recommended alternative medical facilities for elderly people that were centered on the actual requirements and issues faced by the patients. The researchers implemented several forms of machine learning to better meet the fundamental requirements of aged healthcare.

Xu *et al*. [46] provided an outline of federated learning strategies, with an emphasis on those utilized in biomedicine. This article discusses and describes the overall solutions to the empirical challenges, equipment failures, and confidentiality issues that are intrinsic in federated learning with an emphasis on the potential and repercussions for healthcare. Li *et al*. [47] map out machine learning and informatics methodologies and their application in the smart medical sector using bibliometric visualization techniques. The investigation draws attention to countries that are engaged in doing too much work on the subject while also monitoring the key research areas, funding agencies, and emerging hotspots of research. Furthermore, it outlines significant challenges and prospective research objectives related to the application of machine learning and data mining techniques within the healthcare sector. Siddiqui *et al*. [48] utilized a deep learning framework to estimate the phases of breast cancer, and one of the techniques they used was data fusion. They utilized a decision-based fusion approach to get a higher degree of precision using the presented methodology.

Since the introduction of artificial intelligence, it has been common practice to use AI applications to aid in illness diagnosis, enhance disease prognosis, and reduce patient treatment periods. Dai *et al*. [49] converted the challenge of hospitalization prediction into an issue of supervised categorization, which led to a considerable set of possible savings in medical spending budgets. Tariq *et al*. [50] created a method based on multi-dimensional fusion and artificial intelligence to predict the severity of COVID-19. Sedik *et al*. [51] identified the coronavirus by using convolutional methods. Additionally, they devised a method for the extraction of features to increase the amount of information that was available. Qayyum *et al*. [52] proposed a strategy based on clustered FL for interpreting medical visual input at the edge. This would allow remote medical facilities to benefit from heterogeneous information while preserving patients' privacy. Brisimi *et al*. [53] used FL to address scattered sparse Support Vector Machine problems, which allowed them to provide accurate therapy predictions for individuals suffering from cardiovascular diseases. However, the above-mentioned centralized training systems involve the summation of medical information of different individuals in a central database, which involves tremendous difficulties due to confidentiality issues related to the data. In contrast, federated learning is a decentralized system, and offers collaborative learning with all personal data kept in its original form. Therefore, it may offer a private approach towards the amalgamation of different types of healthcare data on edge devices. This architecture may be thought of as a combination model that combines elements from both the distributed and centralized learning perspectives. The past few years have seen hundreds of scholarly articles appear that discuss the application of federated learning within the domain of smart healthcare [54-59]. Table **1** provides a comparative review between prior research efforts and the suggested framework, highlighting the key advancements introduced in this study.

3. PROPOSED METHODOLOGY

XGBoost is a machine learning algorithm for information analysis that systematizes tasks for information analysis and gives useful information. Machine learning can be used to find out how much energy is being used, keep track of services, and plan transportation operations, among other things [60]. Datasets in healthcare networks can be modified using diverse machine-learning techniques; therefore, any inaccuracies in the data should be easy to address. Networks demand data consistency. Any data inaccuracies in the proposed system will be disregarded. This study uses an entirely new method of forecasting and diagnosing diseases. In this research, the main objective is to assess the performance of a system model constructed with the application of machine

Table 1. A comparative review of existing literature and the suggested framework.

Authors	Data	Model	Decision Making	Fused Decision Making	Healthcare 5.0 Paradigm	Use of FL
[42]	Medical Records	Yes	Yes	No	No	No
[43]	Medical Records	Yes	Yes	Yes	No	No
[44]	Medical Records	Yes	Yes	Yes	No	No
[45]	Medical Records	Yes	Yes	No	No	No
[47]	Medical Records	No	No	No	No	No
[48]	Medical Records	Yes	Yes	Yes	No	No
Suggested Framework	Sensor data	Yes	Yes	Yes	Yes	Yes

learning and produced for the most precise adapted predictions in the health management field.

Federated learning, as a result of its several advantages for learning distributed and sensitive data, has become extremely popular lately. It is very attractive for this reason. Instead of pooling data from different places or using typical data gathering and replication strategies, federated learning can be used for the development of a worldwide model, which would be hosted on a centralized server [61-66]. Actual data stays safe within an organization. This way, the organizations gain insight without necessarily merging their data. The term "federated learning" refers to a method in which several locations act cooperatively to train a single global model. The procedure of building a global model using training data from many sources is referred to as federated learning. This approach does not include the direct exchange of information. This guarantees that the confidentiality of patient information is maintained across all sites.

Using federated learning, which is maintained by a centralized server, several different companies or institutions work together to solve an issue involving machine learning. As a consequence of this, a central server will take care of hosting and optimizing a deep learning model [67-69]. The training of the model involves distributing itself over remote centralized data centers, hospitals, and other types of medical facilities that may fall under this category. This is done while ensuring that data localization is maintained at each of these sites. During the entirety of the training process, no information from any of the participants will be traded or sent to another individual. In conventional deep learning, the data are not sent to a single server; rather, the server preserves a universally shared framework that is available to all organizations. After that, every institute constructs its model based on the information collected from patients. Simultaneously, each organization will utilize the model's error gradient to interact with the server. Once the central server gathers feedback from all participants, the global model is updated based on the established criteria. Because of the stated criteria, the model can judge how well the answer was done, and it will only contain information that contributes to the model's overall success. Consequently, input from organizations reporting unsatisfactory or anomalous outcomes may be disregarded. This approach generates a singular federated learning cycle, which is then performed continuously until the global model is learned. Fig. (**1**) depicts FL's architecture.

The techniques of data fusion integrate the information from several sensors to provide observations that are more exact than those that could be produced by a single sensor working alone. Information extraction refers to the practice of getting data from a variety of sources that are not necessarily connected, but then integrating that data to have the biggest possible impact. Because a security framework makes use of several network security sensors, it can be interesting to advance a picture that is broad of the overall state of security maintained by a dynamic security system. In addition, it could be challenging to effectively control the devices if they are scattered across a vast region. To get the most out of the model and provide analysis with a comprehensive system protection condition, it is essential to combine the findings of several sensors in a way that is both effective and intelligent. In contrast, the information might give a better level of consistency regarding its reliability if it originates from various sources. This is because the data is obtained from a variety of places.

Sensors are an important component of the Internet of Things because they play an important role in the evaluation of intelligent healthcare systems, ecosystems, and individuals [70]. These include the following gadgets: Computers process the data generated by sensors in a wide variety of fields, including healthcare, cameras, and interactivity. For instance, to regulate a patient's heart rate, the sensor connected to the heart monitor collaborates with the healthcare infrastructure to perform its function. The application layer is a component of the IoMT communication framework that can include various elements like wearable technology, closed-loop systems, and other comparable devices.

The application of machine learning computational methods can improve the intelligence of federated learning-based systems. Using machine learning algorithms based on distributed blockchain technology can improve data privacy. XGBoost can be used to improve understanding and accelerate the rate at which comprehension is obtained through the exchange of more knowledge. The technology and network architecture required to set up a decentralized blockchain platform are provided by it [71]. Within the scope of this study, we evaluate the SVM and XGBoost deployment architecture, which is a complex system. The most effective use of this technology is the gathering of intelligence from a variety of information sources, such as sensors, mobile devices, and IoMT systems, among others. The information that may be gained by applying these methodologies is utilized in the development of intelligent applications. Nevertheless, the proposed approach is applied in the analytical process to analyze and anticipate real-time data [72].

In the process of creating research data, common problems, including data duplication, missing data parameters, bugs, and noise, are mitigated. The machine learning methods are useful in situations in which access to only a limited subset of a data collection is necessary. The architecture is designed to accommodate a diverse range of implementations in a variety of domains, such as the identification and prevention of fraudulent activity. The federated learning-based architecture is comprised of three primary elements: an application layer, an information framework, smart contracts, and the machine learning framework, and concentrates on the edge of the Internet of Medical Things (IoMT). As can be seen in Fig. (**2**), the suggested system uses several hidden layers, hidden neurons, and different activation methods to achieve optimal performance in the healthcare monitoring system. The approach described splits data analysis into three independent steps: data collection, preparation, and evaluation. The evaluation layer consists of prediction and performance sub-layers. For analytical reasons, data is painstakingly collected from sensors and controllers. This data is handled in its raw form by the data gathering layer. To rectify any discrepancies during the preprocessing phase, a thorough data purification and training protocol is executed.

Clinical data collected from IoMT devices often includes various types of noise, incompleteness, and inaccuracies. To address these challenges, our preprocessing layer primarily

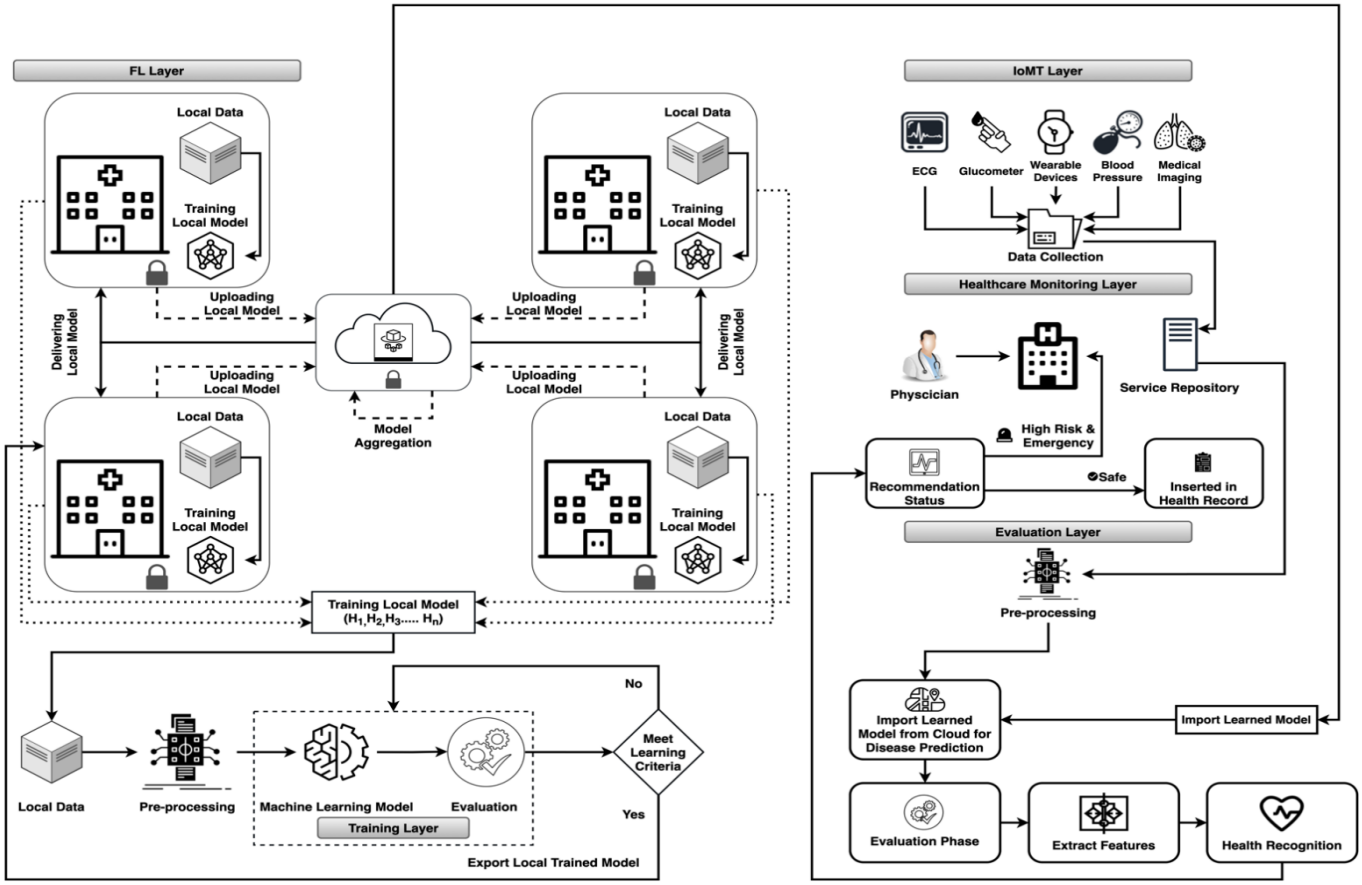

Fig. (2). A healthcare monitoring system utilizing federated learning for healthcare 5.0. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

utilizes moving average and normalization techniques due to their simplicity, effectiveness, and minimal computational overhead. However, alternative methods such as Median Filtering, Kalman Filters, and data interpolation were also evaluated. Empirical tests revealed that moving average and normalization provided the optimal balance between accuracy improvement and computational efficiency for our real-time federated healthcare monitoring context.

The utmost priorities of this method are the following:

- Implementing the machine learning algorithms to develop the most effective strategy for healthcare monitoring.
- An intelligent algorithm that can discover and monitor healthcare in patients as well as determine the severity of a patient's condition is offered in this research.
- To maintain the required level of medical care, it is necessary to integrate IoMT and make improvements to its existing applications. In addition, a federated learning strategy has been incorporated into the model to make it work more effectively. To improve the efficiency of the system, more testing is performed utilizing illness datasets, and a newly learned layer is added to a previously trained framework.
- To protect the confidentiality of the patient's information, it is necessary to keep records of the local data.
- To assess a trained model's efficacy by testing and performance evaluation on a real-time dataset. Utilizing federated learning, an assessment of the prediction system's implementation may be carried out. Additionally, the results of the model are compared to the results of other machine learning models.
- To stop the transmission of diseases that cause chronic illness in people, and to reduce the risk of such illnesses.
- To evaluate the efficacy of the suggested technique using a variety of datasets.
- In addition, do tests and analyses on the datasets using a variety of machine learning methods to validate the suggested strategy.

The research will offer a new perspective on diverse deployments (such as healthcare data sharing) by utilizing the most current phases of technical development, which will be reflected in a comprehensive overview of technical advancements pertinent to federated learning-based frameworks. In this research, we offer an intelligent algorithm for healthcare monitoring across individuals as its primary goal.

The operation of the suggested model is explained in the following way;

In this paper, hospitals will initially train a local model in the training phase of the IoMT framework and then transfer

it to a centralized server, where it will be accessible as a global model for all IoMT devices. The three stages of the process are the application layer, the preprocessing layer, and the sensor layer. For the preprocessing layer to have as little noisy data as possible, it has to use moving average and normalization algorithms because the clinical data collected by IoMT devices is sometimes imprecise or incomplete. The application layer that carries the clinical data after the product includes the prediction layer and the performance layer. The prediction layer generates some of its output as metrics like precision and miss rate, which is then utilized by the performance layer to track the performance in the prediction phase. If the evaluation is positive, then the model trained locally gets uploaded to the cloud and works as a global model. However, if the assessment is negative, then a retraining of the model is necessary. The refined model is then disseminated to healthcare institutions. Local data in these communities is then applied in the tweaking thereof, thereby further refining it in the cloud space with no need for sensitive patient information to flow directly between hospitals. This method ensures that the confidential data of the patients are kept in a confidential atmosphere but remains liable for the smooth flow of information from one medical institution to another. Newly updated parameters from each of the participants' organizations are summed up in the cloud, and a new global model is prepared. It is then passed on to all organizations involved in the project. Data fusion with this technique causes information from various sources to merge and produces more reliable and consistent results than would be obtained by just one single source. In a request for access to data, the person will need to appeal in proper form to the central server. Once such permissions are assured to exist, the server allows the nodes to collectively train a shared model of data based on federated learning. In such a way, the process continues until such time that the model has reached acceptable standards. After training the model, the results of data-sharing activities are returned to the requesting party, while at the validation stage, input parameters are passed for evaluation, thus becoming an integral part of the health monitoring activity.

A. Severity Determination

The severity of a patient's condition is determined by the output probabilities generated from the trained predictive model. The model outputs a probability score indicating disease presence and severity, calibrated against clinically validated thresholds. For instance, probabilities closer to 1 indicate higher severity levels, necessitating immediate medical intervention, while lower probabilities reflect mild or early-stage conditions, aiding medical practitioners in prioritizing patient care effectively.

B. Federated Learning Model Training and Validation

We have adopted a federated learning approach to ensure privacy in healthcare data processing. The model training process consists of:

- **Local model training:** Each healthcare device trains a model on local data without sharing raw patient information.
- **Secure model aggregation:** A central server collects and combines model updates from different nodes using differential privacy techniques.
- **Validation:** The final aggregated model undergoes extensive testing using multiple performance metrics (accuracy, precision, recall, and F1-score) to assess its robustness.

C. Security and Privacy Measures

To ensure data security in federated learning, we incorporated the following measures:

- **Homomorphic encryption** to protect data transmission.
- **Secure multi-party computation (SMPC)** to ensure privacy-preserving model updates.

Anomaly detection to prevent adversarial attacks in the network.

Anomaly detection within the proposed framework employs the Isolation Forest algorithm due to its efficiency and high accuracy in identifying anomalous patient health records. Isolation Forest isolates anomalies by randomly selecting a feature and then randomly selecting a split value between the maximum and minimum values of the selected feature. Since anomalies are fewer and significantly different from normal instances, fewer splits are required to isolate anomalies, making Isolation Forest highly effective for real-time anomaly detection. The federated training process was optimized by employing iterative hyperparameter tuning, specifically adjusting learning rates, batch sizes, epochs, and the number of local training rounds. Additionally, we integrated secure aggregation protocols and encryption methods such as homomorphic encryption to optimize security during training. These optimizations resulted in efficient, privacy-preserving training across decentralized nodes.

4. RESULTS

In this study, the SVM and XGBoost method was implemented on the dataset. The results were randomly allotted to the training collection (137 samples) and 30% of the tests (59 samples). The data is examined to diagnose Parkinson's disease. SVM and XGBoost are used to determine whether the patient is positive or negative. Then, investigated an assortment of neurons, counting the stimulus of hidden layers, and dissimilar procedures of active functions. Testing assesses the output of SVM and XGBoost to understand if this method is effective.

Additional evaluation metrics were computed to assess the model's performance. The miss rate was calculated as the ratio of incorrect predictions for each class b b, adjusted over the total true instances for all classes, as shown in **Eq. (1)**:

$$\text{Miss rate} = \frac{\sum_{b=0}^{2}\left({}^{Q_b}/_{S_{z\neq b}}\right)}{\sum_{b=0}^{2}(T_b)} \tag{1}$$

where z=0,1

Similarly, accuracy was calculated as the proportion of correct predictions over total predictions, as shown in **Eq. (2)**:

$$\text{Accuracy} = \frac{\sum_{b=0}^{2}({}^{Q_b}/_{S_b})}{\sum_{b=0}^{2}(Q_b)} \tag{2}$$

In **(Eq. 1)** and **(Eq. 2)**, Q demonstrates the predicted output of Parkinson's disease, and s symbolizes the real output. F_0 andS_0 demonstrate the absence of Parkinson's disease in predictive output and the real output. Q_1 andS_1 signifies there is a Parkinson's disease in prognostic results and factual results individually. ${}^{Q_b}/_{Q_b}$ denotes that prognostic and factual results are parallel. Correspondingly, ${}^{Q_b}/_{S_{z\neq b}}$ symbolizes error, whereby outcomes, both predictive and actual, are altering.

In Table **2**, the training performance of the proposed Parkinson's disease prediction framework using the Support Vector Machine (SVM) algorithm is detailed. Out of 137 training samples, 100 were negative and 37 were positive. The model correctly predicted 92 of the 100 negative cases and 39 of the 37 positive cases, resulting in a high degree of classification precision. This indicates the model's ability to differentiate well between healthy individuals and those potentially afflicted with Parkinson's disease.

Table **3** presents the validation results using the same SVM-based framework on a subset of 59 test samples. Among these, 50 samples were negative, and 9 were positive. The SVM model correctly classified 46 negative samples and 6 positive samples, demonstrating slightly lower performance on the validation set compared to training, yet still maintaining acceptable generalization. This reflects the model's consistency and reliability when evaluated on unseen data.

Table **4** summarizes the quantitative evaluation of the model's performance across training and validation phases

Table 2. Training the SVM-enabled proposed framework for parkinson's disease prediction.

Proposed System for Parkinson's Disease Prediction Model (70% of data for training)			
Total number of records (N = 137)		**Outcome (Q0, Q1)**	
Input	Expected result (S0, S1)	Q0 (Negative)	Q1 (Positive)
	S0 = 100 Negative	92	8
	S1 = 37 Positive	5	39

Table 3. Validation of the proposed framework for the estimation of Parkinson's disease empowered with SVM algorithm.

Proposed System for Parkinson's Disease Prediction Model (30% of data for validation)			
Total number of records (N = 59)		**Outcome (Q0, Q1)**	
Input	Expected result (S0, S1)	Q0 (Negative)	Q1 (Positive)
	S0 = 50 Negative	46	4
	S1 = 9 Positive	3	6

Table 4. Evaluation of the proposed system for parkinson's disease estimation during validation and training empowered with SVM algorithm.

-	**Accuracy**	**Miss Rate**
Training	90.97%	9.03%
Validation	88.14%	11.86%

Table 5. Training of the proposed framework for the estimation of Parkinson's disease empowered with XGBoost

Proposed System for Parkinson's Disease Prediction Model (70% of data for training)			
Total number of records (N = 137)		**Outcome (Q0, Q1)**	
Input	Expected result (S0, S1)	Q0 (Negative)	Q1 (Positive)
	S0 = 100 Negative	98	2
	S1 = 37 Positive	2	42

Table 6. Validation of the proposed framework for the estimation of Parkinson's disease empowered with XGBoost.

Proposed System for Parkinson's Disease Prediction Model (30% of data for validation)			
Total number of records (N = 59)		**Outcome (Q0, Q1)**	
Input	Expected result (S0, S1)	Q0 (Negative)	Q1 (Positive)
	S0 = 50 Negative	47	3
	S1 = 9 Positive	1	8

using two critical metrics: accuracy and miss rate. During training, the SVM model achieved an accuracy of 90.97% and a miss rate of 9.03%, while in validation, the accuracy dropped slightly to 88.14% with an associated miss rate of 11.86%. These results confirm that the model performs robustly during training and maintains commendable predictive power during validation, thereby supporting its application in practical Parkinson's disease prediction scenarios within a federated healthcare monitoring framework.

Table **5** shows the suggested approach that was suggested to be employed for the patients' Parkinson's disease forecasting during the training phase. A total of 137 recordings are utilized during the training process. These records are technically broken down into 100 negative records, 37 positive records, and a combination of the two. It can be observed that the forecasting algorithm accurately anticipates 98 negative records of people who do not have Parkinson's disease, whereas only 2 positive records are expected with any degree of precision. In comparison, 37 records are received under the condition of positive found, of which 42 are properly projected as positive found and 2 are imprecisely predicted as negative results created when a positive result exists. In other words, the condition of positive found yields 37 records.

Table **6** displays the suggested approach that was suggested for the patient's Parkinson's disease diagnosis over the validation phase. There are a total of 59 recordings that are utilized during the training process. These 59 records are officially broken down into 50 records of negative records and 9 records of positive records, respectively. It has been noticed that the system for forecasting accurately projects 47 negative records of people who do not have Parkinson's disease, while inaccurately projecting 3 positive records. In comparison, eight records are collected under the condition of positive findings. Of these, one record is correctly predicted as positive, while the remaining eight records are projected imprecisely as having a negative result created when there is a positive result present.

Table 7. Evaluation of the proposed system for Parkinson's disease estimation during validation and training empowered with the XGBoost algorithm

-	Accuracy	Miss Rate
Training	97.22%	2.78%
Validation	93.22%	6.78%

The evaluation of the suggested technique for detecting Parkinson's disease in terms of its accuracy and its miss rate at the training and validation levels is outlined in Table **7**. It was discovered that the proposed technique for the Parkinson's disease prediction framework in training achieves an

accuracy of 97.22 percent while also having a miss rate of 2.78 percent. In addition, the proposed framework that was presented for the prediction of Parkinson's disease achieved an accuracy of 93.22 % throughout the validation process while simultaneously having a miss rate of 6.78 %. Table **8** exhibits that the XGBoost algorithm achieves more accuracy than the SVM algorithm. Table **9** shows the comparison of the proposed strategy with the existing one that is already in practice because it outperforms others due to its accuracy. The proposed system is performing better than previous techniques used in Parkinson's disease prediction. This discussed SVM and XGBoost algorithm-based Parkinson's disease prediction system addresses the above problem and gives a pretty good solution to the challenge presented above.

Table 8. Evaluation of the proposed system for Parkinson's disease estimation during validation and training empowered with XGBoost

-	-	Accuracy	Miss Rate
SVM	Training	90.97%	9.03%
	Validation	88.14%	11.86%
XGBoost	Training	97.22%	2.78%
	Validation	93.22%	6.78%

Table 9. Assessment of the suggested system architecture for Parkinson's disease estimation using results from past studies.

Authors	Outcome
Sheibani *et al.* [73]	90.6%
Tracy *et al.* [74]	90.1%
Sztahó *et al.* [75]	89.3%
Yaman *et al.* [76]	91.25%
Kuresan *et al.* [77]	95.16%
Proposed Model	97.22%

Fig. (**3**) illustrates the comparative performance evaluation of the proposed system for Parkinson's disease prediction using several key statistical metrics. The x-axis represents a range of evaluation metrics, including Accuracy (Training and Validation), Miss Rate (Training and Validation), Sensitivity (Recall), Specificity, Precision (Positive Predictive Value - PPV), F1-Score, False Positive Value (FPV), and Negative Predictive Value (NPV). Each bar in the chart corresponds to the percentage achieved under these categories. The y-axis denotes the performance percentages, ranging from 0% to 100%, which reflects how well the model performed on each specific metric. As shown, the XGBoost-based model outperformed SVM across most metrics, achieving high accuracy, precision, and F1-score, while maintaining low miss and false positive rates, indicating the robustness of the proposed framework in clinical prediction scenarios.

The dataset used in this study consisted of 137 samples for training and 59 samples for validation. While this dataset provided valuable insights, its relatively small size may impact the model's generalizability. Future studies should incorporate larger datasets to enhance validation reliability and improve the model's performance across diverse healthcare settings.

Further performance evaluation is given in Table **10**, which provides four essential classification metrics: accuracy, precision, recall, and F1-score. These are crucial for understanding the system's diagnostic value in clinical practice. The proposed framework achieves an accuracy of 97.22%, indicating excellent overall performance. The precision (82.98%) reflects the model's reliability in identifying true positives, while the recall (88.64%) highlights its ability to detect most actual positive cases. The F1-score (85.78%), which balances both precision and recall, confirms the model's strong predictive validity. These values collectively suggest that the federated learning model empowered with XGBoost and SVM provides high diagnostic accuracy while preserving privacy in distributed healthcare data environments.

Table 10. Performance metrics.

Metric	Value (%)
Accuracy	97.22
Precision	82.98
Recall	88.64
F1-Score	85.78

5. DISCUSSION

The results clearly indicate that the XGBoost model outperformed the SVM in predicting Parkinson's disease across both training and validation datasets. The XGBoost algorithm achieved superior accuracy (97.22% training, 93.22% validation) compared to SVM (90.97% training, 88.14% validation). This suggests that the boosting mechanism in XGBoost effectively reduces bias and variance, enabling better generalization to unseen data. The effectiveness of the proposed federated learning framework is also evident. Notably, the model maintained data privacy while achieving high diagnostic accuracy, which is crucial in the sensitive domain of healthcare. Furthermore, by employing a decentralized training approach, the model minimized risks associated with centralized data storage and addressed data heterogeneity across institutions.

Despite the high performance, several challenges remain. One concern is the relatively small dataset, which may limit generalizability. Expanding the dataset and testing the model on other diseases will strengthen the framework. Additionally, the computational overhead introduced by federated learning and secure aggregation mechanisms, while manageable in this study, may become significant in larger deployments. Security risks, such as adversarial attacks and communication latency, are additional concerns. The study ad-

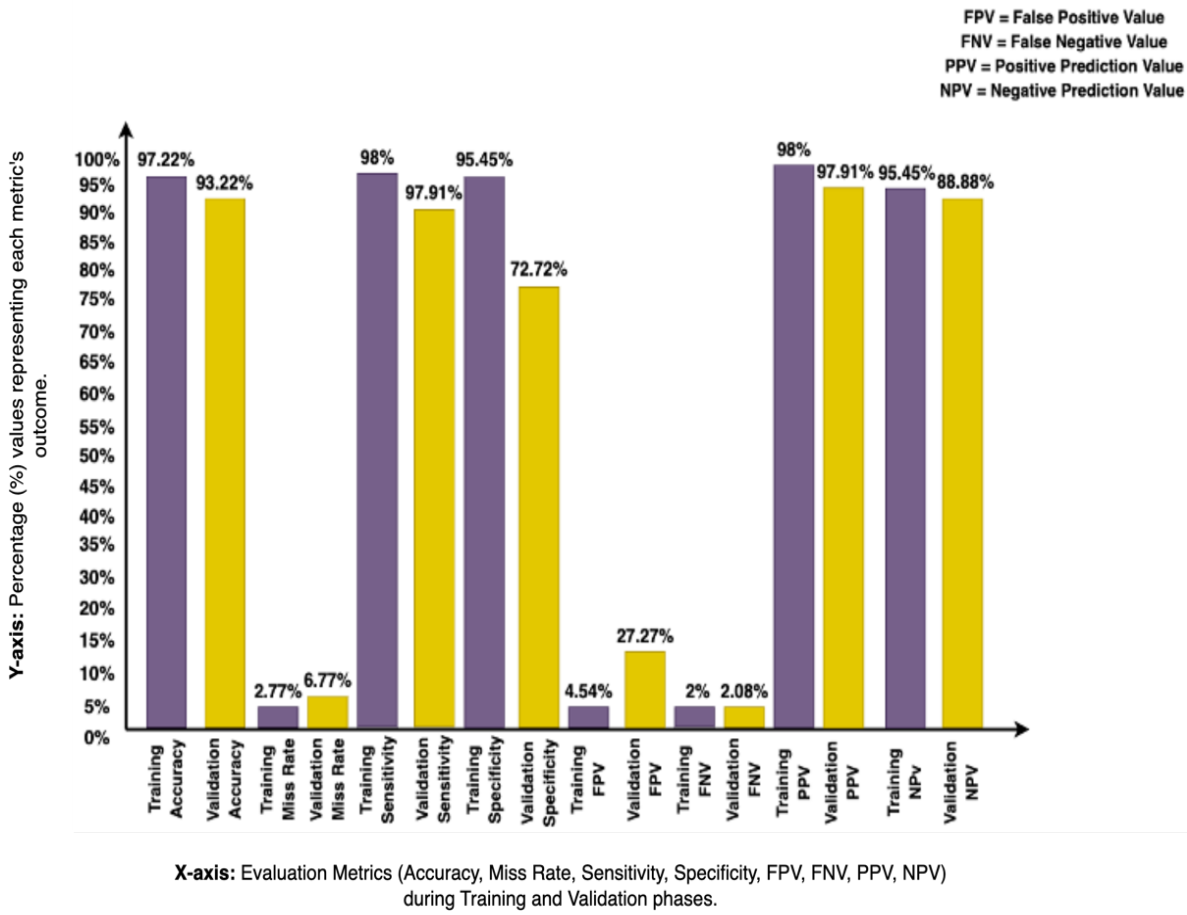

Fig. (3). Performance metrics of the proposed federated learning-based XGBoost model. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

dressed these by integrating anomaly detection and secure multi-party computation. Nevertheless, continued advancements in secure model aggregation and encryption will further enhance robustness.

A. Security Threats and Computational Complexity

While federated learning enhances privacy, it introduces potential security risks, such as:

- **Adversarial attacks:** Malicious nodes may inject false updates into the global model.
- **Computational overhead:** Federated learning requires more computational power compared to traditional centralized models.
- **Communication latency:** Secure model aggregation may introduce minor delays in real-time healthcare applications.

To address these concerns, we implemented anomaly detection methods and optimized the federated training process to minimize overhead.

CONCLUSION

A federated learning-based intelligent approach for improving the prediction performance of healthcare monitoring systems is being investigated. To assess this argument, several different analytic frameworks have been used. The proposed approach described is quite effective. The proposed method has an accuracy rate of 97.22 %. In addition, it is recognized that building a fundamental approach is less costly and quicker. We are delighted with the first findings and have plans to broaden the scope of our investigation by analyzing other data sets shortly. The computational complexity for the proposed system will be limited as the number of hidden layers is increased. Further, research in this regard will focus on the accurate outlining and quantification of the individual constituents. Further selection of varied configurations will be done by increasing the rate of retraining of the learning system. The proposed federated learning-based IoMT model reduces data transmission costs by eliminating the need for centralized data storage. Additionally, human efforts in data handling and security management are significantly reduced, making the system more efficient and scalable for real-world applications.

Currently, our evaluation utilizes a single dataset focused on chronic kidney disease. While results are promising, future validation on multiple real-time datasets from diverse clinical scenarios will enhance the robustness and generalizability of the proposed method. Planned future validations explicitly include datasets covering diabetes, cardiovascular

diseases, and respiratory illnesses, further demonstrating practical utility and versatility.

AUTHORS' CONTRIBUTIONS

It is hereby acknowledged that all authors have accepted responsibility for the manuscript's content and consented to its submission. They have meticulously reviewed all results and unanimously approved the final version of the manuscript.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

Not applicable.

HUMAN AND ANIMAL RIGHTS

Not applicable.

CONSENT FOR PUBLICATION

Not applicable.

AVAILABILITY OF DATA AND MATERIALS

The data and supportive information are available within the article.

FUNDING

None.

CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

ACKNOWLEDGEMENTS

Declared none.

REFERENCES

[1] Yu, Z.; Amin, S.U.; Alhussein, M.; Lv, Z. Research on disease prediction based on improved DeepFM and IoMT. *IEEE Access,* **2021**, *9*, 39043-39054. http://dx.doi.org/10.1109/ACCESS.2021.3062687

[2] Hammad, M.; Iliyasu, A.M.; Elgendy, I.A.; El-Latif, A.A.A. End-to-end data authentication deep learning model for securing IoT configurations. *Human-centric Comput. Inform. Sci.,* **2022**, *12* http://dx.doi.org/10.22967/HCIS.2022.12.004

[3] Babar, M.; Khan, M.S.; Habib, U.; Shah, B.; Ali, F.; Song, D. Scalable edge computing for iot and multimedia applications using machine learning. *Human-centric Comput. Inform. Sci.,* **2021**, *11* http://dx.doi.org/10.22967/HCIS.2021.11.041

[4] Park, J.; Salim, M.M.; Jo, J.H.; Sicato, J.C.S.; Rathore, S.; Park, J.H. CIoT-Net: A scalable cognitive IoT based smart city network architecture. *Human-centric Comput. Inform. Sci.,* **2019**, *9*(1), 29. http://dx.doi.org/10.1186/s13673-019-0190-9

[5] Uddin, Z.; Ahmad, A.; Qamar, A.; Altaf, M. Recent advances of the signal processing techniques in future smart grids. *Hum. Cent Comput. Inf Sci.,* **2018**, *8*(2) http://dx.doi.org/10.1186/s13673-018-0126-9

[6] Yu, B.; Sun, F.; Chen, C.; Fu, G.; Hu, L. Power demand response in the context of smart home application. *Energy.,* **2022**, *240*, 122774. http://dx.doi.org/10.1016/j.energy.2021.122774

[7] Massaoudi, M.; Abu-Rub, H.; Refaat, S.S.; Chihi, I.; Oueslati, F.S. Deep learning in smart grid technology: A review of recent advancements and future prospects. *IEEE Access,* **2021**, *9*, 54558-54578. http://dx.doi.org/10.1109/ACCESS.2021.3071269

[8] Nandy, S.; Adhikari, M.; Khan, M.A.; Menon, V.G.; Verma, S. An intrusion detection mechanism for secured IoMT framework based on swarm-neural network. *IEEE J. Biomed. Health. Inform.,* **2022**, *26*(5), 1969-1976. http://dx.doi.org/10.1109/JBHI.2021.3101686 PMID: 34357873

[9] Zaman, S.; Khandaker, M.R.A.; Khan, R.T.; Tariq, F.; Wong, K.K. Thinking out of the blocks: Holochain for distributed security in IoT healthcare. *IEEE Access,* **2022**, *10*, 37064-37081. http://dx.doi.org/10.1109/ACCESS.2022.3163580

[10] Zarour, M.; Alenezi, M.; Ansari, M.T.J.; Pandey, A.K.; Ahmad, M.; Agrawal, A.; Kumar, R.; Khan, R.A. Ensuring data integrity of healthcare information in the era of digital health. *Healthc Technol Lett,* **2021**, *8*(3), 66-77. http://dx.doi.org/10.1049/htl2.12008 PMID: 34035927

[11] Pallathadka, H.; Mustafa, M.; Sanchez, D.T.; Sekhar Sajja, G.; Gour, S.; Naved, M. IMPACT OF MACHINE learning ON Management, healthcare AND AGRICULTURE. *Mater. Today Proc,* **2023**, *80*, 2803-2806. http://dx.doi.org/10.1016/j.matpr.2021.07.042

[12] Hulsen, T. Explainable Artificial Intelligence (XAI): Concepts and challenges in healthcare. *AI,* **2023**, *652*, 652-666. http://dx.doi.org/10.3390/ai4030034

[13] Dhiman, P.; Bonkra, A.; Sehgal, A.K.; Dhanaraj, R.K. Healthcare trust evolution with explainable artificial intelligence: Bibliometric analysis. *Information,* **2023**, *14*(10), 541. http://dx.doi.org/10.3390/info14100541

[14] Javed, A.R.; Sarwar, M.U.; Beg, M.O.; Asim, M.; Baker, T.; Tawfik, H. A collaborative healthcare framework for shared healthcare plan with ambient intelligence. *Human-centric Comput. Inform. Sci.,* **2020**, *10*(1), 40. http://dx.doi.org/10.1186/s13673-020-00245-7

[15] Tanwar, M.; Khatri, S.K.; Pendse, R. A framework for feature selection using natural language processing for user profile learning for recommendations of healthcare-related content. *Int. J. Bus Anal.,* **2022**, *9*(3), 1-17. http://dx.doi.org/10.4018/IJBAN.292059

[16] Wiens, J.; Shenoy, E.S. Machine learning for healthcare: On the verge of a major shift in healthcare epidemiology. *Clin. Infect. Dis.,* **2018**, *66*(1), 149-153. http://dx.doi.org/10.1093/cid/cix731 PMID: 29020316

[17] Jiang, F.; Jiang, Y.; Zhi, H.; Dong, Y.; Li, H.; Ma, S.; Wang, Y.; Dong, Q.; Shen, H.; Wang, Y. Artificial intelligence in healthcare: Past, present and future. *Stroke Vasc Neurol.,* **2017**, *2*(4), 230-243. http://dx.doi.org/10.1136/svn-2017-000101 PMID: 29507784

[18] Mazhar, T.; khan, S.; Shahzad, T.; khan, M.A.; Saeed, M.M.; Awotunde, J.B.; Hamam, H. Generative AI, IoT, and blockchain in healthcare: Application, issues, and solutions. *Discover Internet of Things,* **2025**, *5*(1), 5. http://dx.doi.org/10.1007/s43926-025-00095-8

[19] Srinivasu, P. N.; Sandhya, N.; Jhaveri, R. H.; Raut, R. From black-box to explainable AI in healthcare: Existing tools and case studies. *Mob Inf Syst.,* 2022 *2022*(1), 1-20.Jun; http://dx.doi.org/10.1155/2022/8167821

[20] Yaqoob, M.M.; Alsulami, M.; Khan, M.A.; Alsadie, D.; Saudagar, A.K.J.; AlKhathami, M.; Khattak, U.F. Symmetry in privacy-based healthcare: A review of skin cancer detection and classification using federated learning. *Symmetry,* **2023**, *15*(7), 1369. http://dx.doi.org/10.3390/sym15071369

[21] Ahmed, D.; Neema, R.; Viswanadha, N.; Selvanambi, R. Analysis and prediction of healthcare sector stock price using machine learning techniques. *Int. J. Inf Syst. Model Des,* **2022**, *13*(9), 1-15. http://dx.doi.org/10.4018/IJISMD.303131

[22] Davenport, T.; Kalakota, R. The potential for artificial intelligence in healthcare. *Future Healthc J.,* **2019**, *6*(2), 94-98. http://dx.doi.org/10.7861/futurehosp.6-2-94 PMID: 31363513

[23] Hasan, T.; Bin Karim, M.F.; Mahadi, M.K.; Nishat, M.M.; Faisal, F. Employment of ensemble machine learning methods for human activity recognition. *J. Healthc Eng.,* **2022**, *2022*, 1-18. http://dx.doi.org/10.1155/2022/6963891 PMID: 36199373

[24] Dhahri, H.; Al Maghayreh, E.; Mahmood, A.; Elkilani, W.; Faisal Nagi, M. Automated breast cancer diagnosis based on machine learning algorithms. *J. Healthc Eng.,* **2019**, *2019*, 4253641. http://dx.doi.org/10.1155/2019/4253641 PMID: 31814951

[25] Tasin, I.; Nabil, T.U.; Islam, S.; Khan, R. Diabetes prediction using machine learning and explainable AI techniques. *Healthc Technol Lett,* **2023**, *10*(1-2), 1-10. http://dx.doi.org/10.1049/htl2.12039 PMID: 37077883

[26] Khan, F.; Tariq, M.; Khan, Z.A.; Ur Rehman, A.; Abbas, S. Cloud-based breast cancer prediction empowered with soft computing approaches. *J. Healthc Eng.,* **2020**, *2020* Article 8017496 http://dx.doi.org/10.1155/2020/8017496

[27] Abbas, S.; Khan, M.A.; Falcon-Morales, L.E.; Rehman, A.; Saeed, Y.; Zareei, M.; Zeb, A.; Mohamed, E.M. Modeling, simulation and optimization of power plant energy sustainability for IoT enabled smart cities empowered with deep extreme learning machine. *IEEE Access,* **2020**, *8*, 39982-39997. http://dx.doi.org/10.1109/ACCESS.2020.2976452

[28] Son, Y.J.; Kim, H.G.; Kim, E.H.; Choi, S.; Lee, S.K. Application of support vector machine for prediction of medication adherence in heart failure patients. *Healthc Inform. Res.,* **2010**, *16*(4), 253-259. http://dx.doi.org/10.4258/hir.2010.16.4.253 PMID: 21818444

[29] Iyortsuun, N. K.; Kim, S. H.; Jhon, M.; Yang, H. J.; Pant, S. A review of machine learning and deep learning approaches on mental health diagnosis. *Healthcare,* **2023**, *11*(3), 285. http://dx.doi.org/10.3390/healthcare11030285

[30] Salas-Zárate, R.; Alor-Hernández, G.; Salas-Zárate, M. D. P.; Paredes-Valverde, M. A.; Bustos-López, M.; Sánchez-Cervantes, J. L. Detecting depression signs on social media: A systematic literature review. *Healthc,* 2022 *10*(2), 291.Feb; http://dx.doi.org/10.3390/healthcare10020291 PMID: 35206905

[31] Khan, M.A. Challenges facing the application of iot in medicine and healthcare. *Int. J. Comput. Inform. Manufact,* **2021**, *1*(1) http://dx.doi.org/10.54489/ijcim.v1i1.32

[32] Albassam, A.; Ijaz, M.; Asghar, M. Integration of blockchain and cloud computing in telemedicine and healthcare. *Int. J. Comput. Sci. Netw Secur.,* **2023**, *23*(6), 59-66.

[33] Khan, A.I. Integrating blockchain technology into healthcare through an intelligent computing technique. *Comput. Mater. Continua,* **2022**, *70*(2) http://dx.doi.org/10.32604/cmc.2022.020342

[34] Housawi, A.A.; Lytras, M.D. Digital transformation from a health professional practice and training perspective. In: *Digital Transformation in Healthcare in Post-Covid-19 Times*; Academic Press, **2023**; pp. 193-204. http://dx.doi.org/10.1016/B978-0-323-98353-2.00011-3

[35] Pagar, K.; Jain, T.; Kumar, H.; Bhardwaj, A.; Handa, R. Medical Appliances energy consumption prediction using various machine learning algorithms. In: *Blockchain and Deep Learning for Smart Healthcare*; WILEY, **2024**. http://dx.doi.org/10.1002/9781119792406.ch14

[36] Aldkheel, A.; Zhou, L. Depression detection on social media: A classification framework and research challenges and opportunities. *J. Healthc Inform. Res.,* **2024**, *8*(1), 88-120. http://dx.doi.org/10.1007/s41666-023-00152-3 PMID: 38273983

[37] Senan, E.M.; Al-Adhaileh, M.H.; Alsaade, F.W.; Aldhyani, T.H.H.; Alqarni, A.A.; Alsharif, N.; Uddin, M.I.; Alahmadi, A.H.; Jadhav, M.E.; Alzahrani, M.Y. Diagnosis of chronic kidney disease using effective classification algorithms and recursive feature elimination techniques. *J. Healthc Eng.,* **2021**, *2021*, 1-10. http://dx.doi.org/10.1155/2021/1004767 PMID: 34211680

[38] Rehman, A.; Abbas, S.; Khan, M.A.; Ghazal, T.M.; Adnan, K.M.; Mosavi, A. A secure healthcare 5.0 system based on blockchain technology entangled with federated learning technique. *Comput. Biol. Med.,* **2022**, *150*, 106019. http://dx.doi.org/10.1016/j.compbiomed.2022.106019 PMID: 36162198

[39] Mohanta, B.; Das, P.; Patnaik, S. Healthcare 5.0: A paradigm shift in digital healthcare system using artificial intelligence, IOT and 5G communication. *International Conference on Applied Machine Learning (ICAML),* Bhubaneswar, India, 25-26 May 2019, pp. 191-196 http://dx.doi.org/10.1109/ICAML48257.2019.00044

[40] Mbunge, E.; Muchemwa, B.; Jiyane, S.; Batani, J. Sensors and healthcare 5.0: Transformative shift in virtual care through emerging digital health technologies. *Global Health. J.,* **2021**, *5*(4), 169-177. http://dx.doi.org/10.1016/j.glohj.2021.11.008

[41] Bhavin, M.; Tanwar, S.; Sharma, N.; Tyagi, S.; Kumar, N. Blockchain and quantum blind signature-based hybrid scheme for healthcare 5.0 applications. *J. Inform. Security Applications,* **2021**, *56*, 102673. http://dx.doi.org/10.1016/j.jisa.2020.102673

[42] Du, X.; Chen, B.; Ma, M.; Zhang, Y. Research on the application of blockchain in smart healthcare: Constructing a hierarchical framework. *J. Healthc Eng.,* **2021**, *2021*, 1-13. http://dx.doi.org/10.1155/2021/6698122 PMID: 33505644

[43] Ihnaini, B.; Khan, M.A.; Khan, T.A.; Abbas, S.; Daoud, M.S.; Ahmad, M.; Khan, M.A. A smart healthcare recommendation system for multidisciplinary diabetes patients with data fusion based on deep ensemble learning. *Comput. Intell. Neurosci.,* **2021**, *2021*(1), 4243700. http://dx.doi.org/10.1155/2021/4243700 PMID: 34567101

[44] Nasr, M.; Islam, M.M.; Shehata, S.; Karray, F.; Quintana, Y. Smart healthcare in the age of AI: Recent advances, challenges, and future prospects. *IEEE Access,* **2021**, *9*, 145248-145270. http://dx.doi.org/10.1109/ACCESS.2021.3118960

[45] Khan, M.F.; Ghazal, T.M.; Said, R.A.; Fatima, A.; Abbas, S.; Khan, M.A.; Issa, G.F.; Ahmad, M.; Khan, M.A. An iomt-enabled smart healthcare model to monitor elderly people using machine learning technique. *Comput. Intell. Neurosci.,* **2021**, *2021*(1), 2487759. http://dx.doi.org/10.1155/2021/2487759 PMID: 34868288

[46] Xu, J.; Glicksberg, B.S.; Su, C.; Walker, P.; Bian, J.; Wang, F. Federated learning for healthcare informatics. *J. Healthc Inform. Res.,* **2021**, *5*(1), 1-19. http://dx.doi.org/10.1007/s41666-020-00082-4 PMID: 33204939

[47] Li, Y.; Shan, B.; Li, B.; Liu, X.; Pu, Y. Literature review on the applications of machine learning and blockchain technology in smart healthcare industry: A bibliometric analysis. *J. Healthc Eng.,* 2021 *2021*, 9739219.Aug; http://dx.doi.org/10.1155/2021/9739219 PMID: 34426765

[48] Yamin Siddiqui, S.; Naseer, I.; Adnan Khan, M.; Faheem Mushtaq, M.; Ali Naqvi, R.; Hussain, D.; Haider, A. Intelligent breast cancer prediction empowered with fusion and deep learning. *Comput. Mater. Continua,* **2021**, *67*(1), 1033-1049. http://dx.doi.org/10.32604/cmc.2021.013952

[49] Dai, W.; Brisimi, T.S.; Adams, W.G.; Mela, T.; Saligrama, V.; Paschalidis, I.C. Prediction of hospitalization due to heart diseases by supervised learning methods. *Int. J. Med. Inform.,* **2015**, *84*(3), 189-197. http://dx.doi.org/10.1016/j.ijmedinf.2014.10.002 PMID: 25497295

[50] Tariq, A.; Celi, L.A.; Newsome, J.M.; Purkayastha, S.; Bhatia, N.K.; Trivedi, H.; Gichoya, J.W.; Banerjee, I. Patient-specific COVID-19 resource utilization prediction using fusion AI model. *NPJ Digit. Med.,* **2021**, *4*(1), 94. http://dx.doi.org/10.1038/s41746-021-00461-0 PMID: 34083734

[51] Sedik, A.; Iliyasu, A.M.; Abd El-Rahiem, B.; Abdel Samea, M.E.; Abdel-Raheem, A.; Hammad, M.; Peng, J.; Abd El-Samie, F.E.; Abd El-Latif, A.A. Deploying machine and deep learning models for efficient data-augmented detection of COVID-19 infections. *Viruses,* **2020**, *12*(7), 769. http://dx.doi.org/10.3390/v12070769 PMID: 32708803

[52] Qayyum, J.Q.; Abd El-Latif, A.A. Collaborative federated learning for healthcare: Multi-modal covid-19 diagnosis at the edge. *arXiv,* **2021**, arXiv:2101.07511v1. http://dx.doi.org/10.48550/arXiv.2101.07511

[53] Brisimi, T.S.; Chen, R.; Mela, T.; Olshevsky, A.; Paschalidis, I.C.; Shi, W. Federated learning of predictive models from federated Electronic Health Records. *Int. J. Med. Inform.,* **2018**, *112*, 59-67. http://dx.doi.org/10.1016/j.ijmedinf.2018.01.007 PMID: 29500022

[54] Medjahed, H.; Istrate, D.; Boudy, J.; Baldinger, J.L.; Dorizzi, B. A pervasive multi-sensor data fusion for smart home healthcare monitoring. *IEEE International Conference on Fuzzy Systems,* Taipei, Taiwan, 27-30 June 2011 , pp. 1466-1473 http://dx.doi.org/10.1109/FUZZY.2011.6007636

[55] Jithish, J.; Alangot, B.; Mahalingam, N.; Yeo, K.S. Distributed anomaly detection in smart grids: A federated learning-based approach. *IEEE Access,* **2023**, *11*, 7157-7179. http://dx.doi.org/10.1109/ACCESS.2023.3237554

[56] Naeem, A.; Anees, T.; Naqvi, R.A.; Loh, W.K. A comprehensive analysis of recent deep and federated-learning-based methodologies for brain tumor diagnosis. *J. Pers Med.,* **2022**, *12*(2), 275. http://dx.doi.org/10.3390/jpm12020275 PMID: 35207763

[57] Ng, D.; Lan, X.; Yao, M.M.S.; Chan, W.P.; Feng, M. Federated learning: A collaborative effort to achieve better medical imaging models for individual sites that have small labelled datasets. *Quant Imaging Med. Surg.,* **2021**, *11*(2), 852-857. http://dx.doi.org/10.21037/qims-20-595 PMID: 33532283

[58] Ma, Z.; Zhang, M.; Liu, J.; Yang, A.; Li, H.; Wang, J.; Hua, D.; Li, M. An assisted diagnosis model for cancer patients based on federated learning. *Front Oncol.,* **2022**, *12*, 860532. http://dx.doi.org/10.3389/fonc.2022.860532 PMID: 35311106

[59] Truhn, D.; Tayebi Arasteh, S.; Saldanha, O.L.; Müller-Franzes, G.; Khader, F.; Quirke, P.; West, N.P.; Gray, R.; Hutchins, G.G.A.; James, J.A.; Loughrey, M.B.; Salto-Tellez, M.; Brenner, H.; Brobeil, A.; Yuan, T.; Chang-Claude, J.; Hoffmeister, M.; Foersch, S.; Han, T.; Keil, S.; Schulze-Hagen, M.; Isfort, P.; Bruners, P.; Kaissis, G.; Kuhl, C.; Nebelung, S.; Kather, J.N. Encrypted federated learning for secure decentralized collaboration in cancer image analysis. *Med. Image Anal.,* **2024**, *92*, 103059. http://dx.doi.org/10.1016/j.media.2023.103059 PMID: 38104402

[60] Rehman, A.; Athar, A.; Khan, M.A.; Abbas, S.; Fatima, A.; Atta-ur-Rahman, ; Saeed, A. Modelling, simulation, and optimization of diabetes type II prediction using deep extreme learning machine. *J. Ambient Intell. Smart Environ.,* **2020**, *12*(2), 125-138. http://dx.doi.org/10.3233/AIS-200554

[61] Tan, Y.N.; Tinh, V.P.; Lam, P.D.; Nam, N.H.; Khoa, T.A. A transfer learning approach to breast cancer classification in a federated learning framework. *IEEE Access,* **2023**, *11*, 27462-27476. http://dx.doi.org/10.1109/ACCESS.2023.3257562

[62] Yang, Q.; Yang, J.; Tan, C.; Shi, H. Federated learning with privacy-preserving and model IP-right-protection. *Mach Intell. Res.,* **2023**, *20*(1), 1-17. http://dx.doi.org/10.1007/s11633-022-1343-2

[63] Su, Z.; Wang, Y.; Luan, T.H.; Zhang, N.; Li, F.; Chen, T.; Cao, H. Secure and efficient federated learning for smart grid with edge-cloud collaboration. *IEEE Trans Industr Inform.,* **2022**, *18*(2), 1333-1344. http://dx.doi.org/10.1109/TII.2021.3095506

[64] Ogier du Terrail, J.; Leopold, A.; Joly, C.; Béguier, C.; Andreux, M.; Maussion, C.; Schmauch, B.; Tramel, E.W.; Bendjebbar, E.; Zaslavskiy, M.; Wainrib, G.; Milder, M.; Gervasoni, J.; Guerin, J.; Durand, T.; Livartowski, A.; Moutet, K.; Gautier, C.; Djafar, I.; Moisson, A.L.; Marini, C.; Galtier, M.; Balazard, F.; Dubois, R.; Moreira, J.; Simon, A.; Drubay, D.; Lacroix-Triki, M.; Franchet, C.; Bataillon, G.; Heudel, P.E. Federated learning for predicting histological response to neoadjuvant chemotherapy in triple-negative breast cancer. *Nat. Med.,* **2023**, *29*(1), 135-146. http://dx.doi.org/10.1038/s41591-022-02155-w PMID: 36658418

[65] Li, Y.; Wei, X.; Li, Y.; Dong, Z.; Shahidehpour, M. Detection of false data injection attacks in smart grid: A secure federated deep learning approach. *IEEE Trans Smart Grid,* **2022**, *13*(6), 4862-4872. http://dx.doi.org/10.1109/TSG.2022.3204796

[66] Wen, M.; Xie, R.; Lu, K.; Wang, L.; Zhang, K. FedDetect: A novel privacy-preserving federated learning framework for energy theft detection in smart grid. *IEEE Internet Things J.,* **2022**, *9*(8), 6069-6080. http://dx.doi.org/10.1109/JIOT.2021.3110784

[67] Zhang, Y.; Suleiman, B.; Alibasa, M.J.; Farid, F. Privacy-aware anomaly detection in IoT environments using fedgroup: A group-based federated learning approach. *J. Netw Syst. Manage,* **2024**, *32*(1), 20. http://dx.doi.org/10.1007/s10922-023-09782-9

[68] Abbas, S.; Khan, M.A.; Al Ghamdi, M.A.; Kim, S.-H.; Bilal, M.; Masood, K. Fused weighted federated deep extreme machine learning based on intelligent lung cancer disease prediction model for healthcare 5.0. *Int. J. Intell. Syst.,* **2023**, *2023*, 2599161. http://dx.doi.org/10.1155/2023/2599161

[69] Singh, P.; Masud, M.; Hossain, M.S.; Kaur, A.; Muhammad, G.; Ghoneim, A. Privacy-preserving serverless computing using federated learning for smart grids. *IEEE Trans Industr Inform.,* **2022**, *18*(11), 7843-7852. http://dx.doi.org/10.1109/TII.2021.3126883

[70] Sakib, S.; Fouda, M.M.; Fadlullah, Z.M.; Nasser, N.; Alasmary, W. A proof-of-concept of ultra-edge smart iot sensor: A continuous and lightweight arrhythmia monitoring approach. *IEEE Access,* **2021**, *9*, 26093-26106. http://dx.doi.org/10.1109/ACCESS.2021.3056509

[71] Khan, M.A.; Abbas, S.; Rehman, A.; Saeed, Y.; Zeb, A.; Uddin, M.I.; Nasser, N.; Ali, A. A machine learning approach for blockchain-based smart home networks security. *IEEE Netw,* **2021**, *35*(3), 223-229. http://dx.doi.org/10.1109/MNET.011.2000514

[72] Haider, A.; Adnan Khan, M.; Rehman, A.; Rahman, M.U.; Seok Kim, H. A real-time sequential deep extreme learning machine cybersecurity intrusion detection system. *Comput. Mater. Continua,* **2021**, *66*(2), 1785-1798. http://dx.doi.org/10.32604/cmc.2020.013910

[73] Sheibani, R.; Nikookar, E.; Alavi, S.E. An ensemble method for diagnosis of Parkinson's disease based on voice measurements. *J. Med. Signals Sens.,* **2019**, *9*(4), 221-226. http://dx.doi.org/10.4103/jmss.JMSS_57_18 PMID: 31737550

[74] Tracy, J.M.; Özkanca, Y.; Atkins, D.C.; Hosseini Ghomi, R. Investigating voice as a biomarker: Deep phenotyping methods for early detection of Parkinson's disease. *J. Biomed. Inform.,* **2020**, *104*, 103362. http://dx.doi.org/10.1016/j.jbi.2019.103362 PMID: 31866434

[75] Sztaho, D.; Valalik, I.; Vicsi, K. Parkinson's disease severity estimation on hungarian speech using various speech tasks. *10th International Conference on Speech. Technology and Human-Computer Dialogue (SpeD 2019),* **2019**. Timişoara, Romania, 10-12 October 2019, pp. 1-6 http://dx.doi.org/10.1109/SPED.2019.8906277

[76] Yaman, O.; Ertam, F.; Tuncer, T. Automated Parkinson's disease recognition based on statistical pooling method using acoustic features. *Med. Hypotheses,* **2020**, *135*, 109483. http://dx.doi.org/10.1016/j.mehy.2019.109483 PMID: 31954340

[77] Kuresan, H.; Samiappan, D.; Masunda, S. Fusion of WPT and MFCC feature extraction in Parkinson's disease diagnosis. *Technol Health. Care,* **2019**, *27*(4), 363-372. http://dx.doi.org/10.3233/THC-181306 PMID: 30664511